

Part 1 — Literature Review & Novelty Assessment

Structural Fingerprints of Chain-of-Thought Perturbation — CoT_Proposal.pdf

Prepared for SWE 4606 / CSE 4610 Design Project · Systems and Software Lab (SSL), IUT

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PART 1 — LITERATURE REVIEW & NOVELTY ASSESSMENT

Headline for your supervisor

Verdict: (ii) **Partially covered.** Your *perturbation taxonomy* has been substantially scooped (a March 2026 paper uses three of your four types on GSM8K). Your *instrument* — a typed reasoning graph parsed from an already-generated CoT — has been built and validated by others, more ambitiously than you propose. But **nobody has joined the two:** no paper measures graph-structural change in verification steps as a function of perturbation type, and no paper validates a CoT parser on perturbed text. Two of your six claimed contributions survive intact; the rest need rewording.

Below, papers are grouped by your requested angles. Everything named here I verified against arXiv/ACL/OpenReview listings; where I could not verify a detail I say so.

(a) CoT perturbation / robustness studies

△ Closest match on taxonomy — read this first

Fragile Thoughts: How Large Language Models Handle Chain-of-Thought Perturbations — Ashwath Vaithinathan Aravindan & Mayank Kejriwal, 2026. arXiv:2603.03332.

The paper evaluates 13 models on a structured taxonomy of five CoT perturbation types — MathError, Unit-Conversion, Sycophancy, SkippedSteps, and ExtraSteps — injected into the reasoning chain on mathematical reasoning tasks. It uses GSM8K and a partial-trace completion task: the model receives a question plus the first few steps of the solution, with perturbations injected at specific points, and must generate the remainder.

Overlap with you — severe, on three fronts. Their *MathError* $\approx$ your arithmetic error. Their *Sycophancy* (a sentence asserting “The problem author thinks...” followed by a false equation) $\approx$ your authority statement. Their *SkippedSteps* $\approx$ your skipped step. Their generation setup (GSM8K, prefix-conditioned continuation, filtering to problems the model gets right) is nearly identical to your §9.2. Only your *contradictory step* has no counterpart.

Where you still differ — and it’s the important part. Their entire dependent variable is accuracy. They define robustness as the ratio of completion accuracy on the perturbed trace to accuracy on the clean trace. Crucially, they *notice* exactly the failure-mode dissociation your §1.1 is built on, but only anecdotally: they contrast a model that never acknowledges the injected error and propagates it, against one that explicitly labels it a typo

and recomputes, against a third that opens with “Let’s carefully review the partial solution and correct any mistakes” but never names the error — which they note has implications for CoT interpretability. That third case is your “noticed it, attempted to verify, verified incorrectly” bucket, observed by hand in a single example. **Your framework is the systematic measurement of what they saw by eye.** That is a genuinely strong framing for your Related Work.

Two numbers from this paper you must plan around (see red flags in Part 2):

- On GSM8K, most large models are near ceiling. Average accuracy drops were 18% for MathError, 9% for UnitConversion, but only ~3% for Sycophancy, ~2% for SkippedSteps, and ~0.4% for ExtraSteps.
- Small models show the effects. Llama 3.1 8B-Instruct dropped 35% under MathError and 17% under Sycophancy; Ministral 3B dropped 58% and 15% respectively. Your 8B-class choice is defensible; anything larger will floor out.

⚠ Closest match on measuring verification behaviour under perturbation

Are Reasoning LLMs Robust to Interventions on their Chain-of-Thought? — Alexander von Recum, Leander Gırrbach, Zeynep Akata (Helmholtz Munich / LMU / TUM), 2026. arXiv:2602.07470.

They introduce a controlled framework that perturbs a model’s own CoT at fixed timesteps, with seven interventions grouped as benign, neutral, and adversarial, applied across nine open-weight reasoning models on math, science, and logic tasks. The adversarial set includes an incorrect reasoning continuation, a fabricated mathematical fact, and an unrelated CoT opening; segmentation splits traces on double newlines, and after each intervention the same model resumes its own chain.

This is the paper that most threatens your novelty claim, because it already measures verification behaviour under perturbation. They classify each of the 20 sentences following an intervention as expressing doubt or not, and find that doubt spikes immediately after intervention with magnitude depending on intervention type — benign interventions produce the smallest increase, while neutral and adversarial ones trigger strong self-questioning. Paraphrasing shows the opposite pattern, reducing doubt below the 0.153 baseline and producing the most consistent drop in final correctness across all interventions.

How you differ. Their “doubtfulness” is a *flat proportion over a fixed 20-sentence window*. It has no graph, no reachability, no notion of what a check points back to, and no node matching between baseline and perturbed chains. Your Check Reach and Perturbation-to-Check Distance are not expressible in their framework. But their result means **you can no longer claim that “no existing work asks whether distinct perturbation types leave distinct traces” (your §3, Measurement gap)**. They asked, and answered, with a non-structural readout. Your gap must be narrowed to: *no work asks whether the traces are distinguishable in graph structure, and whether structure adds anything over a flat cue count*.

This also directly endangers your ablation A5. Their doubt classifier achieved 93.75% accuracy and Cohen’s κ of 0.8742 against a majority of four human annotators, comparable to the human pairwise average of 0.8385. A cheap sentence-level verification detector already separates intervention types. A5 is now not a hypothetical adversarial baseline — it is a published, working method. Plan for the real possibility that A5 wins.

Other verified work in this space

Paper	What it does	Relation to you
Dissociation of Faithful and Unfaithful Reasoning in LLMs — Yee, Li, Tang, Jung, Paturi, Bergen, COLM 2024, arXiv:2405.15092	Investigates how LLMs recover from errors in CoT, finding evidence of unfaithfulness when models reach correct answers despite invalid reasoning text, and identifying factors that shift recovery behaviour — models recover more from obvious errors and in contexts with more evidence for the correct answer.	Your intellectual ancestor for “different failure modes hide behind one accuracy number.” Cite it in §1.1. They use hand-annotated recovery categories, not graph structure.
Measuring Faithfulness in Chain-of-Thought Reasoning — Lanham et al., 2023, arXiv:2307.13702	Measures faithfulness by deliberately adding mistakes to the chain of thought, finding wide variation in post-hoc reasoning with larger models tending to be less faithful. Also truncates and paraphrases.	The original accuracy-only CoT-intervention design. Correctly described in your §2.1.

Paper	What it does	Relation to you
Language Models Don't Always Say What They Think — Turpin, Michael, Perez, Bowman, NeurIPS 2023, arXiv:2305.04388	Shows CoT explanations are heavily influenced by biasing features added to model <i>inputs</i> — e.g. reordering multiple-choice options so the answer is always (A) — which models systematically fail to mention, causing accuracy drops up to 36% on 13 BIG-Bench Hard tasks.	Your §2.1 misdescribes this. See “Citation corrections” below.
Can Language Models Perform Robust Reasoning in CoT Prompting with Noisy Rationales? — Zhou, Tao, Zhu, Luo, Wang, Han, NeurIPS 2024	Noisy rationales supplied as in-context examples. Per von Recum et al., this work does not modify the model’s own reasoning trace.	Adjacent; different intervention surface.
BadChain: Backdoor Chain-of-Thought Prompting — Xiang et al., ICLR 2024	A targeted backdoor attack injecting subtle semantic violations into reasoning chains.	Security framing of the same mechanism. Worth one citation line.
Reasoning Robustness of LLMs to Adversarial Typographical Errors — Gan et al., EMNLP 2024	Single-character typos reduce Mistral-7B GSM8K accuracy from 43.7% to 38.6%.	Surface-level, not step-level. Contrast, not competition.

(b) Graphs as post-hoc diagnostic (not execution engine)

This is where your “instrument gap” claim needs the most revision. Several groups have built the parser you describe.

△ Closest match on the instrument

ReasoningFlow: Semantic Structure of Complex Reasoning Traces — Jinu Lee, Sagnik Mukherjee, Dilek Hakkani-Tür, Julia Hockenmaier, 2025. arXiv:2506.02532. Extended as **ReasoningFlow: Discourse Structures for Understanding LLM Reasoning Traces**, arXiv:2606.05402 (June 2026).

Each trace is segmented into nodes — contiguous, non-overlapping snippets from clauses to sentences — and assigned one of 8 node classes based on semantic role, such as Plan, Reasoning, and Reflection. Edges connect these into a labelled DAG with three main edge types (reasoning, planning, evaluation) and 14 fine-grained labels. The schema surfaces deductive/inductive reasoning, verification, backtracking, and proof-by-contradiction as subgraph patterns. Edges always flow from earlier to later steps, mirroring left-to-right autoregressive information flow; sentences are the default node unit, split further when two clauses have distinct functional roles.

Overlap. This *is* the “prior graph-parsing diagnostic design” your §2.3 describes as a straw man — except it’s real, published, and better resourced than your reconstruction of it. Your §7.2’s claim that three categories are “the minimal set” is now a claim you must defend against a validated 8-class scheme, not against an unnamed predecessor.

Where you still differ — and this survives. Their manual validation covers 31 math/physics/chemistry questions with QwQ-32B-Preview traces at temperature 0, annotated by four authors with two per trace, reaching Krippendorff’s α of 0.8851 for node classification and 0.9193 for edge detection. **Every one of those traces is clean.** They never apply the parser to deliberately corrupted text and never test whether agreement holds there. Your §12.2 perturbed-vs-unperturbed κ comparison and your §12.3 gate remain unoccupied ground. Also relevant to your H3: they report that mechanistically measured causal step-to-step dependencies do not align with language-level discourse relations — a caution for anyone treating a text-parsed edge as a causal dependency, which you should absorb into your Limitations.

Other verified post-hoc graph work

Graph-Based Chain-of-Thought Pruning for Reducing Redundant Reflections in Reasoning LLMs — 2026, arXiv:2604.05643. Splits the raw CoT into steps using keyword triggers, then performs iterative LLM-driven graph construction where each node is labelled either *progress* (advances reasoning) or *review* (reflective behaviour), converting each linear CoT into a DAG with explicit dependency edges. Their progress/review split is functionally your Compute-vs-Check distinction. Their goal is pruning redundant verification for efficiency, not diagnosis — but they demonstrate the two-way labelling works. **Flag: this weakens your claim that the three-way scheme is a novel design choice.**

Thought Anchors: Which LLM Reasoning Steps Matter? — Paul C. Bogdan, Uzay Macar, Neel Nanda, Arthur Conmy, NeurIPS 2025. arXiv:2506.19143. Three attribution methods: black-box counterfactual importance from 100 rollouts conditioned on a sentence or a semantically different replacement; aggregated

attention patterns identifying “broadcasting” sentences that receive disproportionate attention via receiver heads; and a causal attribution method. The paper labels sentences per a taxonomy and releases a directed acyclic graph among sentences. Several of their eight functional sentence roles map onto ReasoningFlow labels — plan generation, fact retrieval, deduction/active computation, example testing, final answer emission. Their resampling method is orders of magnitude more expensive than yours; your cost-bounding argument holds against them specifically.

Graph of Verification (GoV) — Jiwei Fang, Bin Zhang, Changwei Wang, Jin Wan, Zhiwei Xu, AAAI 2026. arXiv:2506.12509. Models reasoning as a DAG, enforces topological order so premises are verified before conclusions, verifies each unit against previously validated antecedents, and terminates at the first detected error to enable precise fault localization. Verification *by* graph, not diagnosis of verification behaviour. Different goal, easily distinguished.

Mapping the Minds of LLMs: A Graph-Based Analysis of Reasoning LLM — 2025, arXiv:2505.13890. Clusters long verbose CoT outputs into semantically coherent reasoning steps, then constructs directed reasoning graphs capturing contextual and logical relations.

From Chains to DAGs: Probing the Graph Structure of Reasoning in LLMs — 2026, arXiv:2601.17593. Finds DAG structure is meaningfully encoded in LLM representations, with recoverability peaking in intermediate layers and varying by node depth, edge span, and model scale. Internal-representation probing, not text parsing.

■ Naming collision you should fix

Verifying Chain-of-Thought Reasoning via Its Computational Graph (CRV) — Zheng Zhao, Yeskendir Koishkenov, Xianjun Yang, Naila Murray, Nicola Cancedda (FAIR at Meta), Feb 2026. arXiv:2510.09312. Hypothesizes that attribution graphs of correct CoT steps possess distinct structural fingerprints from incorrect steps, trains a classifier on structural graph features, and finds the signatures are highly domain-specific. They conclude that correct and incorrect reasoning leave distinct structural fingerprints, and that targeted interventions on transcoder features can correct faulty reasoning.

Their graphs are *internal circuit attribution graphs*, not text graphs — genuinely different. But they own the phrase “structural fingerprints” in exactly this context, published in a top lab, six months before your proposal. **Recommend retitling** to something like “*Verification Survival Under Chain-of-Thought Perturbation*” to avoid a reviewer assuming you’re a text-level knockoff of a Meta paper. Cite CRV as a complementary mechanistic counterpart.

(c) Self-verification steps and their fragility

Self-Verification Dilemma: Experience-Driven Suppression of Overused Checking in LLM Reasoning — 2026, arXiv:2602.03485. Analyses rollouts from Qwen3-8B, QwQ-32B, gpt-oss-20b and DeepSeek-R1-Distill-Qwen-7B, finds a persistently high ratio of unnecessary to necessary rechecks, and validates an LLM-based reflection classifier through human evaluation on a random subset of 1,000 reflection steps. They inject a strong suppression signal asserting epistemic closure and observe the effect on rechecking behaviour.

This is the single most direct precedent for your authority-suppression hypothesis. They deliberately suppress verification via injection and measure the result. They do it to save tokens; you do it to diagnose. Cite it, and note explicitly that their intervention is designed to suppress, whereas yours tests whether an ordinary false-authority statement suppresses as a side effect.

A Mechanistic View of Authority Hierarchy in LLM Sycophancy — 2026, arXiv:2607.00415. Finds authority-induced erasure of correct answer representations is not fully reversible through chain-of-thought reasoning; models produce fluent, confident traces that arrive at the wrong answer, often constructing post-hoc justifications for the hinted choice rather than reasoning independently. Supports your authority prediction from the mechanistic side.

Trust Me, I’m an Expert: Decoding and Steering Authority Bias in Large Language Models — Mammen, Joswin, Venkitachalam, 2026, arXiv:2601.13433. Reported to find that models exhibit authority bias in CoT reasoning, with higher-authority endorsements disproportionately swaying outputs, and that incorrect expert endorsements increase both error rates and confidence in wrong answers. *I verified this exists and is cited by Fragile Thoughts, but did not read the full text — verify before citing directly.*

■ Your authority prediction may be backwards

Your Table 2 and Table 12 predict authority statements produce **low** Check Survival. Two findings push against this:

- von Recum et al. find that adversarial interventions including hallucinated facts trigger strong doubt signals, with doubt levels slightly higher in traces that eventually reach the correct answer.
- Fragile Thoughts finds frontier models largely resist sycophantic influence, with only a 3% average accuracy drop, and that failures often take the form of “perturbation-induced reinterpretation” — the model does not adopt the false equation but is destabilised into reinterpreting the problem entirely.

That last one is a *different structural signature* than the one you predict, and arguably a more interesting one: not “verification vanishes” but “verification survives and audits the wrong thing.” I’d add it as a named alternative outcome rather than let a reviewer point it out.

(d) Cheap deterministic typed-graph parsing validated on perturbed text

Nothing found. This is your cleanest open ground.

What exists: ReasoningFlow validates on clean traces with Krippendorff’s α ; von Recum et al. validate their doubt classifier on a synthetic set of 200 GPT-generated doubtful sentences and 200 non-doubtful sentences sampled from CoT traces — note that is a purpose-built balanced set, not a stratified perturbed-vs-unperturbed comparison of the real corpus; the Self-Verification Dilemma paper validates its reflection classifier on 1,000 sampled steps without splitting by condition.

No paper I found computes labeller agreement **separately on perturbed and unperturbed subsets and treats the difference as a reported statistic**, nor gates its own reporting on that difference. Your §12.2–12.3 is defensible as novel. This is the contribution to lead with.

(e) Verification of citations already in CoT_Proposal.pdf

Your ref	Status	Correction needed
[1] Wei et al., CoT Prompting, NeurIPS 2022	✓ Real, correct	—
[2] Cobbe et al., GSM8K, arXiv:2110.14168	✓ Real, correct	—
[3] Besta et al., Graph of Thoughts, AAAI 2024	✓ Real — arXiv:2308.09687, first authors Besta & Blach, Gerstenberger corresponding	Add arXiv ID
[4] Turpin et al. 2023	✓ Real — arXiv:2305.04388, NeurIPS 2023	⚠ Misdescribed. Your §2.1 lumps [4,5] as “inserting arithmetic mistakes, truncating, paraphrasing, or adding misleading cues.” Turpin adds biasing features to model <i>inputs</i> , e.g. reordering multiple-choice options. Only Lanham [5] intervenes on the chain. Split the sentence.
[5] Lanham et al. 2023, arXiv:2307.13702	✓ Real, correctly described	—
[6] Geva et al., StrategyQA, TACL 2021	✓ Real	—
[7] Cohen 1960 κ; [8] McFadden 1974; [9] Dunn 1964	✓ All real, standard	—
[10] “GraphReason/SaGoT and GraphIC (to be cited with full details)”	✓ All three real — fill this in now	GraphReason: Lang Cao, <i>Enhancing Reasoning Capabilities of LLMs through A Graph-Based Verification Approach</i> , arXiv:2308.09267. Constructs a reasoning graph over multiple CoT outputs, merging coincident steps and using a GNN for verification. GraphIC: Jiale Fu et al., arXiv:2410.02203. Constructs thought graphs — directed, node-attributed graphs modelling reasoning steps and dependencies — for in-context example retrieval. SaGoT: <i>Self-attention-based Graph-of-Thought for Math Problem Solving</i> , Findings of ACL 2025.

Follow-ups from these lines you should also check: Besta’s group and Yao et al.’s *GoT: Effective Graph-of-*

Thought Reasoning (Findings of NAACL 2024) — I confirmed the latter exists but did not read it; it is execution-engine work and should not threaten you.

Do any of them already do what you propose? No. GoT/GraphIC/SaGoT/GraphReason are all execution or retrieval or verification engines — your §2.2 characterisation is accurate and holds. The threats come from ReasoningFlow (instrument), Fragile Thoughts (taxonomy), and von Recum et al. (verification-under-perturbation), none of which you currently cite.

Verdict

(ii) **Partially covered.**

Already exists: (1) The perturbation taxonomy — Aravindan & Kejriwal (2026) apply MathError, Sycophancy, and SkippedSteps to GSM8K prefix-continuation across 13 models. (2) The typed reasoning-graph parser with a verification node class and human-validated agreement — Lee et al.’s ReasoningFlow (2025/2026), at higher granularity than your three-way scheme. (3) The finding that verification behaviour changes measurably and differently by intervention type — von Recum et al. (2026), via a flat doubt-sentence proportion.

Genuinely open: (1) *Graph-structural* measurement of verification under perturbation — survival, reach, and reachability distance from the injected step — as opposed to a positional or proportional count. (2) Content-first node matching across a baseline/perturbed graph pair, which no perturbation study needs because none of them build graphs. (3) Parser validation on perturbed text specifically, with per-subset κ and a hard reporting gate. Nothing in the literature does this, and given that von Recum’s paper applies an LLM classifier to adversarially altered traces without checking whether the classifier degrades there, the gap is live and consequential.

Net: proceed, but reposition. You are no longer proposing the first structural look at CoT perturbation. You are proposing the first *graph-structural* one, and the first *validated-on-the-text-it-is-applied-to* one. Contributions 2, 5, and 6 in your §21 stand as written. Contributions 1 and 3 need rewording. Ablation A5’s premise has changed and needs its own defence.

Ready-to-drop differentiation statement

Three lines of recent work bound this proposal. First, controlled CoT-perturbation studies have converged on a taxonomy close to ours: Aravindan and Kejriwal (2026) inject mathematical errors, false expert assertions, and step omissions into GSM8K reasoning prefixes across thirteen models, and von Recum et al. (2026) apply seven benign, neutral, and adversarial interventions at fixed positions in a model’s own chain. Both report accuracy or answer-recovery as the outcome; von Recum et al. additionally count doubt-expressing sentences in a fixed window after the injection, establishing that verification behaviour responds differently to different intervention types. Second, post-hoc graph parsers for reasoning traces already exist: ReasoningFlow (Lee et al., 2025) segments traces into eight semantic node classes joined by fourteen edge labels, achieving Krippendorff’s α above 0.88 against human annotators, and related work labels steps as progress versus review to prune redundant reflection. Third, mechanistic approaches such as CRV (Zhao et al., 2026) recover structural error signatures from internal attribution graphs rather than from the text. We adopt none of these wholesale. From the first line we keep the intervention design but replace the readout, because a count of doubt sentences in a positional window cannot say whether a check is downstream of the damage in the reasoning or merely later on the page — a distinction only reachability in a graph can draw. From the second line we keep the graph but strip it back deliberately, to three node types and two edge types with only heuristic-flagged edges escalated to a model call, because the richer schemas were validated on clean traces and offer no evidence that they remain reliable on corrupted ones. That last point is the gap this work is built around: every published parser we surveyed reports agreement on unperturbed text and then, in the perturbation setting, applies the same labeller to adversarially altered input without re-checking it. An unvalidated parser applied to corrupted text cannot distinguish a change in the model’s reasoning from a change in the parser’s behaviour. We therefore validate labelling separately on perturbed and unperturbed subsets, report the difference as a statistic in its own right, and gate all structural claims on it.